CSE563 — Unit III Numerical Worksheet

Probability and Statistics

40 questions • CO3 • Student Worksheet

Standard normal table extract

z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.58
$\Phi(z)$	0.6915	0.8413	0.9000	0.9332	0.9500	0.9750	0.9772	0.9950

$t_{0.025,\,3} = 3.182$, $t_{0.025,\,8} = 2.306$, $t_{0.025,\,9} = 2.262$, $t_{0.025,\,15} = 2.131$, $t_{0.025,\,24} = 2.064$

$e^{-1} = 0.3679$, $e^{-1.5} = 0.2231$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$, $e^{-4} = 0.0183$

Instructions

Attempt all 40 questions and show your working. Round probabilities to four decimal places. For every hypothesis test, state H_0 and H_1, the test statistic, the decision, *and a conclusion in the context of the question.* Refer to the *Unit III Numerical Guide* for a worked example of each problem type.

A. Basic and Compound Probability (Q1–Q10)

Q1. Two fair dice are rolled. Find the probability that (a) the sum is 8, (b) the sum is at least 10, (c) both dice show the same value.

Q2. A 4-character password uses letters A–Z with repetition allowed. Find the probability that a randomly generated password (a) has all distinct characters, (b) contains at least one repeated character.

Q3. From a standard 52-card deck, 3 cards are drawn without replacement. Find the probability that all three are hearts.

Q4. A committee of 5 is selected from 6 men and 4 women. Find the probability it contains exactly 3 women.

Q5. $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$, $P(A \mid B)$ and determine whether A and B are independent.

Q6. A bag has 5 red and 7 blue balls. Two are drawn without replacement. Find the probability that (a) both are red, (b) they are of different colours.

Q7. A system has three independent components each working with probability 0.9. Find the probability that (a) all three work, (b) at least one fails.

Q8. A disease affects 2% of a population. A test has sensitivity 95% and specificity 90%. Find the probability that a person who tests positive actually has the disease.

Q9. Factories A, B and C supply 25%, 35% and 40% of components, with defect rates 2%, 1% and 3%. A component is found defective. Find the probability it came from each factory, and verify the posteriors sum to 1.

Q10. Two machines produce 60% and 40% of items, with 3% and 5% defective respectively. Find (a) the overall defect rate, (b) the probability a defective item came from machine 2.

B. Probability Distributions (Q11–Q20)

Q11. A fair coin is tossed 8 times. Find the probability of exactly 5 heads, and the mean and variance of the number of heads.

Q12. 20% of items from a production line are defective. In a sample of 10, find the probability of (a) exactly 3 defective, (b) no defectives, (c) at least 2 defective.

Q13. A factory finds that 1% of its chips are faulty. In a batch of 300, use the Poisson approximation to find the probability of at most 2 faulty chips.

Q14. Emails arrive at a server at 4 per hour. Find the probability of (a) exactly 2 in an hour, (b) none in 30 minutes, (c) at least 1 in 15 minutes.

Q15. Defects occur in a cable at 1.5 per kilometre. Find the probability that a 2 km stretch has exactly 3 defects.

Q16. Heights are normally distributed with $\mu = 165$ cm and $\sigma = 10$ cm. Find (a) $P(X > 175)$, (b) $P(150 < X < 180)$, (c) the height exceeded by only 5% of people.

Q17. Component lifetimes are $N(\mu = 1200, \sigma = 100)$ hours. Find the probability a component lasts (a) more than 1400 hours, (b) between 1100 and 1300 hours. (c) Find the 90th percentile.

Q18. A random variable X has $P(X{=}1) = 0.2$, $P(X{=}2) = 0.5$, $P(X{=}3) = 0.3$. Find $E[X]$, $\mathrm{Var}(X)$ and $E[3X - 1]$.

Q19. In a game, you win Rs.50 with probability 0.2, win Rs.10 with probability 0.3 and lose Rs.20 with probability 0.5. Find the expected gain and state whether the game is fair.

Q20. The number of daily server crashes follows a Poisson distribution with mean 0.5. Find the probability of (a) no crashes in a day, (b) more than 1 crash in a day, (c) no crashes in a 7-day week. *Use* $e^{-0.5} = 0.6065$, $e^{-3.5} = 0.0302$.

C. Frequentist Inference (Q21–Q30)

Q21. A sample of $n = 100$ has $\bar{x} = 52$ and known $\sigma = 10$. Construct 95% and 99% confidence intervals for μ.

Q22. A sample of 25 light bulbs has $\bar{x} = 1050$ hours and $s = 100$ hours. Construct a 95% confidence interval for the mean lifetime.

Q23. In a poll, 420 of 1000 voters support a proposal. Construct a 95% confidence interval for the true proportion.

Q24. How large a sample is needed to estimate a mean to within ± 2 units at 95% confidence, given $\sigma = 15$?

Q25. A manufacturer claims a mean weight of 250 g. A sample of $n = 36$ gives $\bar{x} = 244$ g with known $\sigma = 12$ g. Test at $\alpha = 0.05$ (two-sided).

Q26. A sample of 9 measurements gives $\bar{x} = 10.4$ and $s = 0.6$. Test $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$ at $\alpha = 0.05$.

Q27. A new algorithm is claimed to reduce mean runtime below 200 ms. A sample of $n = 49$ gives $\bar{x} = 195$ ms, $\sigma = 21$ ms. Test at $\alpha = 0.05$ (one-sided).

Q28. For a test of $H_0 : \mu = 50$ with $\sigma = 10$, $n = 25$ and $\alpha = 0.05$ (upper-tailed), find the rejection boundary in original units, and compute β and the power when the true mean is 55.

Q29. Explain, with the relevant probabilities, the difference between a Type I and a Type II error for a test conducted at $\alpha = 0.01$ with power 0.85.

Q30. A 95% confidence interval for a mean is $[23.5,\ 28.5]$. State the sample mean and margin of error, and state whether $H_0 : \mu = 30$ would be rejected at $\alpha = 0.05$.

D. Bayesian Inference (Q31–Q34)

Q31. A coin is flipped 15 times giving 11 heads. With a uniform Beta$(1, 1)$ prior, find the posterior distribution, its mean, and the MAP estimate. Compare with the MLE.

Q32. Repeat the previous question with an informative Beta$(10, 10)$ prior. Explain the difference.

Q33. A quality inspector uses a Beta$(2, 8)$ prior for the defect rate. A batch of 50 items contains 8

defectives. Find the posterior mean and compare it with the sample proportion.

Q34. A naive Bayes spam filter has $P(\text{spam}) = 0.4$. The word "win" occurs in 40% of spam and 2% of ham; "invoice" in 5% of spam and 25% of ham. Classify a message containing both words.

E. Regression Analysis (Q35–Q40)

Q35. For $x = 1, 3, 5, 7, 9$ and $y = 3, 7, 10, 16, 19$, compute $\bar{x}$, $\bar{y}$, S_{xx}, S_{yy} and S_{xy}.

Q36. Using the previous question's values, find the least-squares regression line and interpret the slope.

Q37. Compute r and R^2 for the data of Q35, and state what R^2 means here.

Q38. For the fitted line $\hat{y} = 0.75 + 2.05x$, compute the fitted values and residuals for all five points, and verify the residuals sum to zero. Hence find SSE.

Q39. Test whether the slope of the regression in Q36 is significantly different from zero at $\alpha = 0.05$, given $t_{0.025,3} = 3.182$.

Q40. Using $\hat{y} = 0.75 + 2.05x$, predict y at $x = 4$ and at $x = 25$. Comment on the reliability of each prediction.